

Electronic Companion

Accepted Service Adaptation: Finite Continuation Prices and the Value of Memory

Revision R29, September 23, 2026

This companion contains the supporting derivations for the current theorem chain and the exact experimental specification. The complete R24 manuscript and companion, including the earlier neural/value-gradient and certificate-development results, are preserved byte for byte in the separate code/data research archive. They are not additional sections needed to interpret the main theorems. The primary institution in this revision is participation at every review; the ex ante-only master agreement is the case in which the nonroot participation constraints are absent.

EC.1 Finite-Price Experiments and Independent Checks

EC.1.1 The exact design

The production algorithm and its proofs appear in main Sections 4–6. No smoothness assumption or approximation grid is used in these new experiments. The retained sections below give the complete derivations for the broader models in the print appendices; their earlier experimental records are not new R29 observations.

In a generated T -date graph with $S \in \{2, 3\}$ regimes, the root is unique and each later layer has S vertices. Hence $N = 1 + (T - 1)S$. Every nonterminal regime connects to all regimes of the next layer. For current regime index z , successor index j , and seed s , the unnormalized probability is $1 + ((z + 2j + s) \bmod 3)$; divide by the sum over successors. Discounting is $19/20$. Using the standard-library pseudorandom generator with the declared seed, draw integer reward numerators from $\{-4, \dots, 44\}$ and divide by 20, curvature numerators from $\{8, \dots, 24\}$ and divide by 10, and payment numerators from $\{8, \dots, 16\}$ and divide by 10. The complete rational primitive instances are stored, so replay does not depend on redrawing them.

Let the date-specific reference tier be $c_t = (7 + (t \bmod 5))/20$. Compute its conditional payment P_v backward. Compute the discounted annuity $A_v = 1 + (19/20) \sum_w p_{vw} A_w$. The cap is $B_v = P_v + ((3v + s) \bmod 7)A_v/100$. The exact root promise is P_0 . The reference table is therefore feasible under every continuation cap. In the unrestricted cases the physical interval is $[0, 1]$; in the fixed-table corridor cases it is $[c_t - 1/4, c_t + 1/4]$. These are two different declared feasible classes, not a claim that the fixed table has already been optimized. Optimized time-only comparisons use the unrestricted primitives and the exact condensed quadratic program in Proposition 6.1.

The main suite consists of the strict-memory example, seven boundary instances, twelve seeded small graphs, nine unrestricted horizon/state configurations, and six fixed-corridor configurations. Every stored instance includes its full primitive graph. The coefficient bit length in the timing record is the maximum numerator or denominator bit length among stored affine response coefficients. Serialized bytes include decimal encodings, not an idealized compressed-memory estimate. Peak resident memory is labeled as the whole sequential generator's peak; it is not represented as isolated configuration memory.

EC.1.2 Response verification is not a sampled feasibility test

The checker does not import the price solver. At each vertex it forms the union of all submitted parent and child knots, both local tier knots, and the submitted cap threshold. Between consecutive points, both sides of the response identity are affine after also accounting for the cap threshold. Equality is checked at every knot, at an interior midpoint of every interval, and on both constant tails. The midpoint and tail checks also verify the reflected implementation and cap complementarity. The input lists must be strictly ordered, continuous at their knots, nonincreasing, and constant on their tails. Thus these exact rational checks certify the entire displayed functions, not only the root query.

A separate forward check reconstructs the discounted mass entering every public/price pair. It verifies local interval feasibility, the actual conditional payment recursion, each continuation cap, root equality, stationarity, sign restrictions, and complementarity. Finally it compares the total reward with main equation (14). The verifier rejects a missing child state or a mass mismatch. The certificate needs no claim that a floating-point optimizer has identified an exact active set.

The independent tree audit is deliberately different. It unfolds only the fourteen small test graphs, builds the conditional subtree-incidence matrix directly, and checks matrix KKT conditions on that extensive form using rational arithmetic. It then uses a feasibility linear program followed by SLSQP with target objective tolerance 10^{-12} . The separate numerical comparison is accepted only with objective discrepancy below 10^{-7} and constraint violation below 10^{-8} . Those thresholds do not relax the rational optimality certificate. No production complexity count includes a small-test tree as though it were required for execution.

For the optimized table comparison, a conventional primal active-set method supplies rational candidates. The second checker reconstructs reach weights, conditional payment coefficients, objective curvature, all cap rows, and the root row independently from the primitive graph. It verifies the exact primal–dual bracket rather than trusting the active-set iteration count. The shared-design cuts are checked at 54 joint perturbations of table entries and release; each queried value also has its own exact certificate. These are fixed-table oracle tests, not a numerical claim of finite convergence of a continuous outer master.

The two-service check uses discount one and a deterministic serial expansion of the two local coordinates. The extra intermediate cap is redundant. The transformation preserves the one scalar commitment and the original node’s continuation cap; it does not split the problem into independently budgeted service blocks. The general vector-tier corollary is proved in the main paper, not inferred from this one check.

EC.1.3 Memory calculations and reproducibility

In the strict-memory example, a public-only terminal tier c satisfies $0 \leq c \leq 1/4$. Write the intermediate tiers as y and $1 - 2c - y$. The cap in regime A gives $0 \leq y \leq 1/4 - c$. For fixed c , the derivative of reward with respect to y is $1/2 - c - y \geq 1/4$, so the optimal y is $1/4 - c$. The resulting derivative with respect to c is $5/2 - 2c > 0$. Hence $c = 1/4$, $y = 0$, and the other tier is $1/2$. Direct substitution gives $13/32$, versus the finite-price optimum $27/32$. Their difference is $7/16$.

For the memory-frontier family, caps are $b_j = j/(k + 1)$, with $k = 2, 3, 4, 5, 6, 8, 16, 32$. The checker verifies every dynamic-programming recurrence and the minimizing partition’s reward. For $k \leq 6$, it

additionally enumerates all set partitions, including noncontiguous partitions, and compares their best costs with the ordered solution. Across those five sizes there are 277 partitions. This separates a proof of the ordered-cell reduction from a numerical search that presupposes it.

The scripts and immutable primitive/certificate records distinguish deterministic arithmetic from observed timing. Standard-library rational replay covers the production responses, memory frontier, shared-table certificates, and cut checks. Only the separate numerical tree experiment uses NumPy and SciPy. The build record gives software versions, command outputs, hashes, page counts, and the final source-preservation checks. The historical R24–R28 replays are run unchanged; their outputs are not merged into the new timing table.

EC.2 Identified Stocks and the Participation Reduction

For a public demand regime z , let

$$C_z(S) = cS + h\mathbb{E}(S - D_z)^+ + p_0\mathbb{E}(D_z - S)^+, \quad f_z(S) = \mathbb{E} \min(S, D_z), \quad \Psi_z(S) = C_z(S) - \xi f_z(S),$$

where $\xi > 0$, the stock interval $I_z = [l_z, u_z]$ is compact, and $h + p_0 + \xi > 0$. These symbols describe the physical inventory problem; p_0 is not the tier reward vector p of the main paper.

Proposition EC.2.1 (Stock identification and relaxed commitments). *Suppose s_z uniquely minimizes Ψ_z on I_z . On an exogenous regime process, commitments $F^\pi \geq F^s$ and $C^\pi \leq C^s$ force $S_t = s_{z_t}$ almost surely at every positive-probability review, including randomized policies. Both commitments bind. Writing H_z for the demand distribution function, an interior minimizer satisfies*

$$H_z(s_z-) \leq \frac{p_0 + \xi - c}{h + p_0 + \xi} \leq H_z(s_z).$$

The endpoint conditions are $\Psi'_{z,+}(l_z) \geq 0$ and $\Psi'_{z,-}(u_z) \leq 0$. A strict atomic straddle or a continuously increasing crossing ensures uniqueness; a quantile level in $(0, 1)$ does not alone ensure an interior feasible stock.

If $\Psi_z(S) - \Psi_z(s_z) \geq \alpha|S - s_z|^2$ on all feasible intervals, then

$$F^\pi \geq F^s - \delta_F, \quad C^\pi \leq C^s + \delta_C \quad \implies \quad \mathbb{E}^\pi \sum_t \beta^t |S_t - s_{z_t}|^2 \leq \frac{\delta_C + \xi \delta_F}{\alpha}.$$

A demand density bounded below by $\rho > 0$ throughout the feasible interval is sufficient with $\alpha = (h + p_0 + \xi)\rho/2$.

Proof. The one-sided derivatives are

$$\Psi'_{z,-}(S) = c - p_0 - \xi + (h + p_0 + \xi)H_z(S-), \quad \Psi'_{z,+}(S) = c - p_0 - \xi + (h + p_0 + \xi)H_z(S).$$

Convexity gives the minimization conditions. Each gap $\Psi_z(S) - \Psi_z(s_z)$ is nonnegative. Because the regime process is exogenous, the comparator and the proposed policy have the same discounted regime occupation

probabilities. The expected sum of gaps is $(C^\pi - C^s) - \xi(F^\pi - F^s) \leq 0$. Thus every positive-probability gap is zero and uniqueness identifies stock. Under relaxed commitments the same sum is at most $\delta_C + \xi\delta_F$, and quadratic growth gives the inequality. The density lower bound gives $\Psi_z'' \geq 2\alpha$ almost everywhere; integration with the constrained first-order condition proves growth even when s_z is an endpoint. No smooth-density conclusion is attributed to an atomic instance. $\square$

For several stocks, additive expected costs and a separate service commitment for each coordinate give the same result with $\Psi_z = C_z - \sum_j \xi_j f_{zj}$. Marginal demand correlation does not affect additive expected costs. Product stock sets and jointly feasible coordinate minimizers give a unique vector minimizer. Under coordinate quadratic growth the bound uses the smallest curvature and numerator $\delta_C + \sum_j \xi_j \delta_{F,j}$. Arbitrary coupled stock constraints or a nonadditive service functional require their own minimization argument.

After identification, the remaining expected service benefit at history n is fixed. Customer utility is that benefit minus the conditional net payment. Requiring it to exceed the outside utility yields main equation (2). Nonroot outside opportunities may depend on public history. When they do, their induced caps need not have a regime-only description, and the main recursive-state theorem requires augmenting the state. Keeping only the root constraint recovers the ex ante-only master agreement; it is not evidence for continuation acceptance.

EC.3 Relative Transfer Coordinates

Assume positive scalar payment coefficients and a fixed root payment. For any feasible reference z , put $f_n = -\sum_{i \geq n} c_i(x_i - z_i)$ and $y_n = f_n/c_n$ for nonroot n . At a nonroot node,

$$c_n(x_n - z_n) = \sum_{v \in \text{ch}(n)} f_v - f_n,$$

and at the root the same formula has no f_0 because total payment is unchanged. This is exactly $x - z = By$ with the incidence matrix of Lemma 3.2. Conversely, summing the incidence identity over a subtree cancels its internal transfers and leaves $-c_n y_n$. The maps are therefore mutual inverses, including a comparator on a tier boundary.

The continuation inequality becomes

$$-c_n y_n \leq b_n - \sum_{i \geq n} c_i z_i = \sigma_n.$$

Other rows, bounds, and equalities are substituted without being dropped. With an affine information restriction satisfied by z , $M(x - z) = 0$ becomes $MBy = 0$. A postorder traversal computes every subtree total, and a parent-child pass computes the incidence product, each in order N arithmetic operations. This count assumes fixed-cost scalar arithmetic; rational bit complexity and growth of the history tree are separate issues.

For completeness, the exact quadratic gain in these coordinates is

$$W_\lambda(z + By) - W_\lambda(z) = (B^\top(p - Qz))^\top y - \frac{1}{2}y^\top B^\top QBy - \lambda \sum_a \kappa_a (|(Dz - h)_a + (DBy)_a| - |(Dz - h)_a|).$$

It applies to nonconstant optimized references and permits negative y . Its proof is expansion of the quadratic plus the subtree identity. The value theorem in the main text additionally solves a release problem and identifies the limiting primal and dual events; that extra claim is not obtained by presenting this identity as a new optimization theorem.

EC.4 Restriction-Release Derivation and Event Calculation

EC.4.1 The global frontier

The feasible release graph consists of a fixed compact accepted polyhedron and linear bounds with right sides proportional to t . Taking convex combinations of two graph points shows that V is concave. Monotonicity follows from nesting. For continuity at zero, take any sequence of maximizers as t decreases to zero; compactness gives a convergent subsequence whose limit satisfies the exact restriction. Its value cannot exceed K . The reverse inequality follows from feasibility of the restricted optimizer. Interior continuity follows from concavity. At other endpoints of a finite feasible parameter interval, mixtures with a feasible interior policy provide the needed approximating policies.

One can treat the absolute-value terms by their finitely many sign regions. On a region the objective is a strictly concave quadratic in the tiers, and its sign boundaries are linear constraints. Select an independent basis of active equations and solve its KKT system. Positive definiteness of Q makes the reduced quadratic solution unique and affine in a scalar right-side parameter. There are only finitely many candidate active patterns. A finite refinement at feasibility events and equal-value crossings yields the global continuous piecewise-affine policy and piecewise-quadratic value. Degenerate multipliers may require several adjacent representations of the same policy; this does not justify applying a regular signed-corner formula when its strict margins are zero.

The Lagrangian for the two release bounds subtracts $(\xi^+)^\top (Mx - m_0 - t\omega) + (\xi^-)^\top (-Mx + m_0 - t\omega)$. Holding any optimal multipliers fixed supplies a supporting affine upper bound in t , with slope $\omega^\top (\xi^+ + \xi^-)$. At a differentiability point this directional supporting slope is unique. At a breakpoint the two one-sided slopes can differ. Integrating over finitely many quadratic pieces gives main equation (23).

EC.4.2 Reduced curvature and the affine price path

On a regular cell, write

$$\bar{p} - Qz = R^\top \alpha_0 + M^\top \xi,$$

where R contains the permanent equalities, the currently active accepted bounds, and the zero switching differences. The nonzero switching costs have already been incorporated in $\bar{p}$. Rows of R have different dual

domains: accepted inequality multipliers are nonnegative, permanent equality multipliers are unrestricted, and zero-edge tensions lie in their switching intervals. These domains must be checked separately.

For a prescribed release δ , any cell-feasible displacement has $R(x - z) = 0$ and $M(x - z) = \delta$. Write it as Zv . Its quadratic cost is $v^\top H v / 2$ and its linear gain, apart from the restriction rent, vanishes on the permanent face. Minimizing this quadratic subject to $MZv = \delta$ gives

$$v = H^{-1} Z^\top M^\top S \delta, \quad x = z + P \delta, \quad P^\top Q P = S.$$

Thus the gain is $\xi^\top \delta - \delta^\top S \delta / 2$. The full row-rank assumption ensures that MZ has full row rank and that S is positive definite. Changes of basis Z leave P and S unchanged.

For the signed corner $\delta = t v$, with $v_j = \omega_j \text{sign } \xi_j$, the release multiplier is $\xi(t) = \xi - t S v$. The old face multiplier is

$$\alpha(t) = \alpha_0 - t \dot{\alpha}, \quad R^\top \dot{\alpha} = Q P v - M^\top S v. \quad (1)$$

The last equation has a unique solution because R has independent rows and its right side is orthogonal to $\ker R$: $Z^\top Q P v = Z^\top M^\top S v$. It follows that

$$\bar{p} - Q(z + t P v) = R^\top \alpha(t) + M^\top \xi(t).$$

This equation verifies stationarity throughout the segment. It also tells us when an old participation multiplier vanishes or a zero-edge tension reaches a switching capacity. Merely checking the new policy against the continuation caps would miss these dual events.

Every event has the form $a + t \dot{a} = 0$ or $a + t \dot{a} = \bar{a}$. Include only positive event times in the direction being followed. For an inactive row $A_i x \leq b_i$, the event is $(b_i - A_i z) / (A_i P v)$ if its denominator is positive. Tier bounds and nonzero edge-sign crossings use the same rule. For release component j , its event is $|\xi_j| / [\text{sign}(\xi_j)(S v)_j]$ if this denominator is positive. Old nonnegative multipliers and bounded tensions are handled by (1). The minimum of these event times is t_1 . All regularity margins are strict at zero, so $t_1 > 0$. At the event, the old KKT witness remains valid with an equality at its limiting sign or capacity, establishing inclusion of the endpoint.

If some restriction price is zero, maximizing $\xi^\top \delta - \delta^\top S \delta / 2$ over the release box is still a small strictly concave quadratic program, but its optimum need not be the full signed corner used in main equation (24). If $[R; M]$ is rank deficient, remove redundant restrictions and adjust the release geometry instead of inverting a singular matrix. A nonzero-dimensional accepted face is not silently assumed in the global theorem. These qualifications are why the regular segment is a precise local result embedded in a general frontier.

EC.4.3 Gain targets and contractual localization

On the initial segment, $a = \sum_j \omega_j |\xi_j| > 0$ and $q = v^\top S v > 0$. Since the release multipliers retain their signs, $a - q t \geq 0$ through the segment. Solving $a t - q t^2 / 2 = g$ on its increasing branch gives main equation (26). It is a minimal release because it inverts the maximum attainable accepted value, not the value of an arbitrary feasible policy.

For a subtree n , $\sum_{i \geq n} c_i(x_i(t) - z_i) = t \sum_{i \geq n} c_i d_i$. A positive sum consumes continuation slack and a nonpositive sum does not. Thus only subtrees with positive consumption can be the next cap event. This localizes the first bottleneck but does not say that all eventual expansion is supported on a single cut, nor that the subtree with the smallest numerical slack necessarily binds first. Slack must be divided by its consumption rate. A different release scale, direction, or active cell can change the first subtree.

EC.5 Exact Examples and Comparator Reoptimization

EC.5.1 The two-segment release frontier

In main equation (27), the root multiplier is $-x_0$. On the first segment the child-1 cap multiplier is zero and the multiplier on $x_1 - x_2 \leq t$ is $1/2 - t/2$. The stationarity vector is

$$(-x_0, 1 - x_1, -x_2) = (-x_0, -x_0 + \xi, -x_0 - \xi).$$

Child 1 first reaches $3/4$ at $t = 1/6$. On the second segment its cap multiplier is $3t - 1/2$ and the release multiplier is $3/4 - 2t$. Both are nonnegative precisely through the stated segment. At $t = 3/8$ the release multiplier vanishes and the cap multiplier is $5/8$. Complementarity and primal feasibility establish global optimality because the negative objective is convex. The gain at the first breakpoint is $11/144$, and the final gain is $23/192$. The first policy uses exactly $1/12$ of child-1 continuation slack.

EC.5.2 The U-shaped optimized premium

For $b \in [1/4, 2/3]$ in main equation (29), direct substitution gives

$$J(b) = -3b^2/4 + 7b/4 - 9/16, \quad K(b) = -3b^2 + 4b - 9/8.$$

The full-class multiplier for $x_1 \leq b$ is $7/4 - 3b/2$ and its root multiplier is $-x_0^X$. For the restricted optimizer, the cap multiplier is $4 - 6b$, the root multiplier is $-x_0^Y$, and the equality $x_1 - x_2 = 0$ has multiplier $b - x_0^Y$. These witnesses satisfy stationarity and complementarity. Other bounds have zero multipliers, including at an endpoint where they happen to bind. Subtracting the two objectives gives $9(2b - 1)^2/16$. Subtracting the two cap prices gives its derivative. Hence both optima are genuinely reoptimized, and the U-shape is not an artifact of holding a bad comparator fixed.

EC.5.3 Increasing friction can increase optimized expansion

For main equation (31), the smooth objective is maximized without constraints at $p = (1/2, 3/4, 1/4)$. Every restricted policy satisfies $|x_1 - x_0| + |x_2 - x_0| \geq |x_1 - x_2| = 1/2$. The policy p attains that lower switching cost and the largest possible smooth reward, so it remains the restricted optimum for every nonnegative friction. For $0 \leq \lambda \leq 1/4$, the full optimum has switching tensions $(\lambda, -\lambda)$ and gradient $p - x = (0, \lambda, -\lambda)$. For larger friction the constant policy uses tensions $(1/4, -1/4)$, which remain within the enlarged capacities.

The root multiplier and other bound multipliers can be zero. These are full KKT witnesses; subtracting the two values yields the displayed formula.

EC.6 Budget and Friction Sensitivity Details

For a fixed accepted inequality matrix and equality restriction, the dual objective is affine in the budget vector when its multipliers are held fixed. Polyhedral normal-cone optimality supplies attained multipliers for the strictly convex quadratic plus a finite absolute-value function, even when there is no strictly feasible tier. Thus an optimal multiplier is a supporting supergradient of the value in every feasible budget direction. Along a one-dimensional budget path, the finitely many active representations give a continuous piecewise-quadratic value. Its one-sided derivatives are read from the adjacent pieces. At interior differentiability points, every optimal multiplier has the same product with the budget-path direction.

At a degenerate interior point of the feasible parameter domain, the right derivative of the concave value is the minimum of the products of that direction with its supergradients, equivalently with the projected optimal budget multipliers; the left derivative uses the maximum. At a boundary, only feasible inward directions are relevant, and normals to the feasible parameter domain must be retained in the supporting-price set. Using the adjacent feasible parametric pieces is an equivalent, unambiguous procedure. The difference of two values must use their respective one-sided derivatives. Selecting unrelated multipliers from two degenerate optimizer outputs is not a valid derivative computation.

For friction, compactness and strict concavity make the optimizer continuous. Comparing each optimum under λ and $\lambda + \varepsilon$ sandwiches the value increment between $-\varepsilon T$ at the two optima. Division and continuity give the envelope derivative. Refining the restricted parametric path at every change in accepted active rows and switching signs makes the full-class balance system linear in the friction and its certificate prices. Feasibility of that system is exactly zero expansion. Projection can be obtained by two linear programs for the endpoint frictions on the cell, along with a feasibility check. Empty cells and singleton projections must not be omitted. The count of restricted active cells can be large; this exact threshold construction is not a claim of polynomial complexity or of monotonicity that the optimized restriction need not possess.

EC.7 Primal–Dual Economic Certificates

The recorded computational model uses displacement x from a feasible legacy protocol and the objective

$$F(x) = b_h^\top x - \frac{1}{2} \sum_i d_i x_i^2 - \frac{1}{2} \|Ux\|^2 - \lambda \sum_a \kappa_a (|(Bx + v)_a| - |v_a|),$$

subject to $\ell \leq x \leq u$, $Ax \leq a$, and $Ex = 0$. Here B is the switching matrix, not the relative-transfer matrix in the main tree lemma. Let

$$q_i^*(r) = \max_{\ell_i \leq z \leq u_i} \{rz - d_i z^2/2\}, \quad z = \min\{u_i, \max\{\ell_i, r/d_i\}\}.$$

For arbitrary p , unrestricted η , nonnegative μ , and $|s_a| \leq \lambda \kappa_a$, the bound

$$U(p, s, \mu, \eta) = \sum_i q_i^* ((b_h - U^\top p - B^\top s - A^\top \mu - E^\top \eta)_i) + \frac{1}{2} \|p\|^2 - s^\top v + \mu^\top a + \lambda \sum_a \kappa_a |v_a| \quad (2)$$

is an upper bound on the full optimum. Adding the information equalities to E gives an upper bound on the corresponding restricted optimum.

Indeed, $\|Ux\|^2/2 \geq p^\top Ux - \|p\|^2/2$, and $\lambda \kappa_a |(Bx + v)_a| \geq s_a (Bx + v)_a$. Accepted inequalities give $\mu^\top Ax \leq \mu^\top a$, and equality terms vanish. Maximizing the remaining separable quadratic over the box gives (2). All these steps hold for any feasible displayed prices; no optimizer status is a premise.

The implementation rounds proposals to rational nonroot/group coordinates, reconstructs root payments exactly, and mixes with a strict reduced-coordinate anchor only when needed. For inequality rows $a_j^\top x \leq b_j$ and a strict anchor z , the minimum feasible ray fraction is

$$\alpha = \max \left\{ 0, \max_{j: a_j^\top x > b_j} \frac{a_j^\top x - b_j}{a_j^\top x - a_j^\top z} \right\}, \quad \hat{x} = (1 - \alpha)x + \alpha z.$$

The anchor and proposal share all required equalities, so these remain exact. Rows already satisfied remain satisfied by convexity. The archive retains the rounded proposal's repair fraction. The separate verifier recomputes all rows and both objective bounds using rational arithmetic; it does not accept a feasibility tolerance. Strict feasibility is needed for this specific ray repair, not for the weak-duality bound or the underlying accepted-balance theorem.

For the fixed-amendment class, the root equalities and uniformity constraints imply the singleton zero displacement. We solve this class algebraically, including exact equality multipliers, rather than burden a numerical solver with a redundant zero-dimensional feasible set. For all other classes, the optimizer proposes both a decision and dual prices. A wide rational bracket would be reported as wide; changing the optimizer tolerance cannot itself certify a tighter value.

EC.8 Comparator Cache: Proof and Arithmetic

Let $x_i^* = \arg \max_Y W_{h_i}$ and let ϕ be strongly convex in metric D_0 . Applying the strong-convexity inequality at its constrained minimizer gives

$$W_{h_i}(x_i^*) - W_{h_i}(z_i) \geq \frac{1}{2} \|x_i^* - z_i\|_{D_0}^2.$$

Thus any upper certificate gives the same inequality with right-side bound δ_i . Differences of feasible policies belong to the column space of Z_Y . The induced operator norm gives $\|U(x_i^* - z_i)\| \leq \sqrt{2L_Y \delta_i}$.

For two exact maximizers $x^*(h)$ and $x^*(\tilde{h})$, adding their strong variational inequalities gives

$$\|x^*(h) - x^*(\tilde{h})\|_{D_0}^2 \leq (h - \tilde{h})^\top U(x^*(h) - x^*(\tilde{h})).$$

Use the restricted operator bound on the right and then on the resulting policy difference to obtain $\|Ux^*(h) - Ux^*(\tilde{h})\| \leq L_Y \|h - \tilde{h}\|$. The envelope argument gives $\nabla K = Ux^*$, including nonsmooth switching in ϕ . Integration of this Lipschitz gradient gives the quadratic upper Taylor bound. Substitution of the approximate anchor gradient and its error proves Proposition D.1.

The recorded Z_Y spans the nullspace of both exact root and time-only equalities, and D_0 is the diagonal quadratic curvature. We use the rational trace

$$L_Y = \text{tr}(UZ_Y(Z_Y^\top D_0 Z_Y)^{-1} Z_Y^\top U^\top),$$

which bounds the largest eigenvalue of the positive semidefinite matrix. The coupled resource quadratic contributes additional curvature, so ignoring it here is conservative, not invalid. A singular trace value zero means the reward perturbations are constant over the restricted class; the same inequality holds with zero curvature.

To evaluate the square roots without an unreported floating-point tolerance, for a nonnegative rational r and integer denominator d , find the least integer k satisfying $k^2 \geq rd^2$ and use k/d . Integer square roots and exact comparisons implement this outward rounding. Multiplying the two nonnegative upper roots in main equation (48) is conservative. The independent replay checks the trace matrix, every anchor bracket, the recomputed resource vector, each query envelope, and each offline validation bracket. Dropping the approximate-anchor error term would not in general be safe.

Online evaluation stores only r -dimensional anchors and resource gradients, scalar bounds and curvature, and their errors. Its arithmetic count is independent of the number of history nodes after this cache is built. Exact rational bit complexity, cache construction, storage, acceptance of the proposed policy, and cache misses remain separate costs. The claim is no new *restricted optimization* for a cache evaluation, not zero-cost end-to-end deployment.

EC.9 Radius-Sensitive Outer Comparators

For C_0 containing the common box and all fixed equalities, define the perturbation residual $e_{i,\theta}(x) = (A_\theta)_i x - (b_\theta)_i$. If $|e_{i,\theta}(x)| \leq \eta_i$ for $x \in C_0$, any true-feasible point satisfies $A_i^0 x - b_i^0 \leq \rho \eta_i$. Conversely, an outer-feasible point has true-row violation at most $2\rho \eta_i$. A common strict anchor with margin γ_i therefore yields the mixing fraction in main equation (49). The Lipschitz constant can be chosen uniformly because rewards and gradients are bounded on the compact box and uncertainty set; the absolute-value penalty is Lipschitz there as well.

The historical continuation family has

$$A_{\theta,ji}(\rho) = A_{ji}^0(1 + \rho\theta_3\sigma_i), \quad Cx \leq (1 + \rho\theta_2)a^0, \quad \theta \in [-1, 1]^3.$$

It used $A_j^0 x \leq \rho \sum_i |A_{ji}^0| m_i$, $m_i = \max(|\ell_i|, |u_i|)$, and $Cx \leq (1 + \rho)a^0$. This is already a shrinking outer class, not a radius-independent positive-radius set. Its common strict time-only anchor has negative nonroot displacements and exact reconstructed root displacements; positive continuation coefficients make nonroot

subtree inequalities strict. A sufficiently small common scale makes tier and capacity constraints strict. Continuity on a compact small-radius interval supplies the required uniform margins.

The earlier positive-radius observations were too coarse to diagnose a jump at zero. The new sweep holds the context and interior coefficient direction fixed and reduces the radius to $1/65536$. The measured outer loss then approaches zero. A steep rise before radius $1/16$ is compatible with the analytical bound and with an early change of the active comparator constraints. Neither the new sweep nor this local theorem identifies the uncertainty set as statistically calibrated.

Corollary EC.9.1 (Equality-aware and successive outer refinement). *Replace an unrestricted-box envelope by $\eta_i(C_0) = \sup_{\theta, x \in C_0} |e_{i, \theta}(x)|$. The resulting outer class remains valid and is no larger. More generally, if O_k contains every $Y_\theta(\rho)$, then*

$$O_{k+1} = O_k \cap \{x : A_i^0 x \leq b_i^0 + \rho \sup_{\theta, z \in O_k} |e_{i, \theta}(z)| \text{ for all } i\}$$

also contains every true class and is nested inside O_k .

Proof. For a true-feasible x , write $A_i^0 x - b_i^0 \leq -\rho e_{i, \theta}(x)$ and use the relevant containing set in its support bound. Induction gives the successive statement. $\square$

For a polyhedral uncertainty set and an affine perturbation, maximization over its vertices and a pair of linear programs over the equality-restricted polyhedron compute each envelope. This does not claim that a finite number of refinements recovers the exact union or that their cost is always worthwhile. The recorded radius comparison uses the original box envelope, so its reduction in observed small-radius slack is not credited to an unexecuted refinement.

EC.10 Exact Experimental Specification

EC.10.1 Legacy amendments and the common-instance hierarchy

The legacy primitive file is the R23/R24 63-node, two-service design, retained unchanged. We use the first 24 contexts with radius zero from its recorded certificate file, not a newly selected favorable subset. The five pooling keys for node n , service a , depth t , and binary history string h are respectively (a) , (t, a) , (t, h_{last}, a) , $(t, h_{\text{last two}}, a)$, and (n, a) . The first class pools all amendment coordinates per service; exact root commitments then force zero. All other classes permit distinct date-zero/root amplitudes. The regime annotation is a pooling key on this heterogeneous legacy tree, not a claim that all of its primitive coefficients have a regime-only Markov representation.

For each class we append the corresponding equality rows to the same accepted root constraints. We optimize the original quadratic-plus-switching objective with the same box, continuation, and shared-capacity rows. The rational repair rounds group amplitudes, reconstructs each root from the exact payment equation, and mixes only within that group subspace. The strict anchor is time-only and therefore belongs to every date-refining class. The current source uses OSQP for the two-service legacy proposals and the HiGHS quadratic-programming interface bundled with CasADi for the one-service Markov proposals, with an IPOPT

fallback if the latter raises an exception. The singleton class is solved by exact algebra. Each actual proposal engine and any failed attempt are retained in its record. Development attempts with an over-tight OSQP request stalled on a degenerate canonical instance; they were not counted as completed measurements. The finished evidence contains independently verified brackets for every predeclared instance rather than dropping the difficult case.

EC.10.2 A public-Markov service-tier family

The second family has $T = 6$, 63 binary-tree nodes, one tier, baseline $1/2$, and unconditional weights multiplied by 32: $c_n = 2^{5-t(n)}$. The root regime is zero. A left child has regime -1 and a right child has regime $+1$, with probability $1/2$ each. In displacement coordinates $x_n = \theta_n - 1/2$, the diagonal curvature is $c_n(1 + z_n/4)$, the resource-coupling matrix is zero, and the linear reward is

$$c_n \left(\frac{a}{32} + \frac{z_n b}{32} + \frac{t(n)d}{128} \right) x_n.$$

The switching coefficient is $e/128$, with edge weight c_n , temporal difference $x_n - x_{pa(n)}$, and root installed displacement zero. The root payment is $\sum_n c_n x_n = 0$. Every nonroot continuation has displacement slack

$$s_n = \frac{c_n(6 - t(n))(3 + z_n)}{64}.$$

Dividing by c_n shows that the conditional cap is a function of date and current regime. Thus this family, unlike the heterogeneous legacy annotation, has the Markov primitive description required for the sufficient-state theorem. Its physical tier box is $[0, 1]$, and the same pooling keys now equate actual tiers because the baseline is common.

A pseudorandom generator with seed 250923 draws 24 contexts in the order $a \in \{8, \dots, 24\}$, $b \in \{12, \dots, 32\}$, $d \in \{-6, \dots, 6\}$, $e \in \{1, \dots, 5\}$. Every context is retained. The objective's common weight scaling is why shares within a family, rather than raw values across families, are the primary comparison.

For each adjacent class, the archive stores the exact lower and upper values and the enclosing increment interval. Zero-width or very small true increments are not asserted positive from a difference of rounded approximate values. The mean lower-endpoint increments in the main table are numerical summaries with their stated certificate bracket widths, not exact algebraic identities.

EC.10.3 Fine-radius study and the comparator cache

The radius study uses the first eight nominal legacy contexts and the radii

$$0, 2^{-16}, 2^{-14}, 2^{-12}, 2^{-10}, 2^{-8}, 2^{-6}, 2^{-4}, 2^{-3}, 2^{-2}.$$

It uses a deterministic interior coefficient vector per context. For $i = 0, \dots, 7$, the three numerators over eight are

$$(-5 + 2i) \bmod 13 - 6, \quad (3 + i) \bmod 11 - 5, \quad (7 + i) \bmod 13 - 6.$$

The same vector is used at every radius for that context. Perturbation coefficients and outer sets are reconstructed rationally by affine interpolation of the original model’s radius-zero and radius-1/16 coefficient definitions. This interpolation is exact for the *coefficients*, not an interpolation approximation to optimized values. We directly optimize the true and outer comparator at every recorded radius and audit both.

The cache study fixes the nominal geometry, time-only class, and switching parameter 29/64. Seed 25923 draws four six-dimensional anchors with coordinate numerators in $\{-12, \dots, 12\}$ over 16. Thirty-two queries cycle through these anchors and add independent integer perturbations from $\{-4, \dots, 4\}$ over 256. Each deployed query takes the minimum of four analytical envelopes. The recorded optimizer solves at these queries are solely offline checks on envelope quality. Anchor errors and curvature are computed from certified values, not from optimizer residuals.

EC.10.4 Exact promise-state checks and evidence boundaries

Two three-date examples use tiers $\{0, 1/2, 1\}$ and switching coefficients $1/8$ and $3/8$; a four-date example uses tiers $\{0, 1\}$ and coefficient $1/8$. Demand regimes are equiprobable $-1, +1$, with root regime zero, inherited initial tier $1/2$, undiscounted payment equal to the tier, and reward $(3/4 + z/2)x - x^2/2 - \gamma(x - q)^2$. The continuation cap at a nonroot date is $[3(T - t) + z]/4$. The recursion computes the best reward at every attainable exact root payment and compares the complete dictionary with exhaustive enumeration of all deterministic history policies. There are $2 \cdot 3^7 + 2^{15} = 37142$ enumerated policies before acceptance filtering. All arithmetic uses fractions. These tests concern a genuinely finite-action contract; they make no claim that a continuous promise grid is exact.

The release and capacity examples are checked by explicit rational KKT witnesses. Twenty-four additional random rational cells use dimensions 3, 5, and 7, positive diagonal curvature, independent pooling differences and root equality, and explicit first-event ratios. The random generator constructs known restricted KKT prices and checks the resulting policy, stationarity, and value formula; it is an algebraic stress test, not a sample of operating data. The inherited 64 rational tree-transfer instances are replayed separately.

All historical learned and robust timing records remain unchanged. The new work does not conduct a new neural race, estimate field generalization, or fit a calibrated uncertainty set. The exact certificate calculations validate the displayed designed models and arithmetic; they do not establish journal acceptance or substitute for expert scrutiny of the proofs.

EC.11 Continuous-Promise Certificates and Omitted-State Value

EC.11.1 Exact domains, concavity, and deterministic implementation

We give the additional details behind main Theorem C.2. The theorem concerns the scalar continuous-tier specialization, not the general polyhedron with unrecorded shared resources. For each positive-probability successor, assume its promise interval is nonempty. The image of $[0, 1] \times \prod_j [L_{t+1}(j), U_{t+1}(j)]$ under $ax + \beta \sum_j P_j b_j$ is exactly the interval with endpoints in main equation (35), before its intersection with $(-\infty, \bar{b}_t]$. This follows either from interval addition or continuity on a compact convex set. There are no

missing interior promises. If a successor is infeasible, its positive conditional probability prevents retaining the customer in every continuation. Zero-probability successors are omitted. In the experiment all lower endpoints are zero; the theorem also states the general recursive lower endpoint.

For two policies starting from (q^1, b^1) and (q^2, b^2) at the same date and regime, combine their actions on every future public history with a fixed weight α . Exogenous transition probabilities are unchanged. The inherited tiers at all future histories combine with the same weight, exact promises and cap inequalities remain feasible, and

$$-\ell|\alpha(x^1 - q^1) + (1 - \alpha)(x^2 - q^2)| \geq -\alpha\ell|x^1 - q^1| - (1 - \alpha)\ell|x^2 - q^2|.$$

Together with quadratic concavity this proves the value inequality. The mixture is an actual intermediate tier, not a randomized contract. Compactness gives attainment on the finite tree even when all available continuation slack is exhausted.

For implementation, a stored anchor i contains its current action x_i and, for each successor j , weights λ_{ijk} on child anchors. Require

$$\lambda_{ijk} \geq 0, \quad \sum_k \lambda_{ijk} = 1, \quad \sum_k \lambda_{ijk} q_{jk} = x_i, \quad ax_i + \beta \sum_j P_j \sum_k \lambda_{ijk} b_{jk} = b_i,$$

and

$$l_i \leq g(q_i, x_i) + \beta \sum_j P_j \sum_k \lambda_{ijk} l_{jk}.$$

Given a query (q, b) , solve the lower-envelope mixture linear program in main equation (36), apply $x = \sum_i \alpha_i x_i$, and assign $b_j = \sum_{i,k} \alpha_i \lambda_{ijk} b_{jk}$. The child weights $\sum_i \alpha_i \lambda_{ijk}$ give a feasible mixture at (x, b_j) . Choosing an optimal mixture afresh at that state has at least as large a lower value. Backward induction proves the claimed reward without carrying an unobserved mixture label as additional history. Convex action feasibility is indispensable: with actions $\{0, 1\}$, one period, and payment coefficient 1, exact promise $1/2$ is infeasible even though it is the midpoint of two feasible promises.

EC.11.2 Price planes, weak duality, and uniform error

For a child plane mixture, $\mathcal{V}_{t+1}(j, x, b_j) \leq A_j x + B_j b_j + C_j$. Any convex mixture remains a majorant; its weights must be nonnegative and normalized separately for each successor. For arbitrary η and $|s| \leq \ell$, the parent's objective at a feasible allocation is bounded by

$$sq + \eta b + (r - s - a\eta + \beta \sum_j P_j A_j)x - dx^2/2 + \beta \sum_j P_j \{(B_j - \eta)b_j + C_j\}.$$

Independent maximization over $x \in [0, 1]$ and each exact child interval proves main equation (37). The interval support uses the upper endpoint when $B_j - \eta \geq 0$ and the lower endpoint otherwise. In particular, dropping the child cap from a *lower-policy* construction would be invalid, while enlarging an interval in an upper bound only makes that bound weaker. The clipped quadratic conjugate verifies the entire continuum,

not a finite sample of actions. Approximate numerical prices are permitted because their sign, normalization, and support constants are checked directly. Neither strong duality nor a strict feasible anchor is used for validity.

Let $\mathcal{T}_t$ denote the exact Bellman maximization on the same feasible domains. If $f_j \leq h_j$ and $0 \leq h_j - f_j \leq e_j$ on child domains, then

$$0 \leq \mathcal{T}_t h - \mathcal{T}_t f \leq \beta \sum_j P_j e_j. \quad (3)$$

To see this, compare the objectives at each common feasible action–promise allocation, then take maxima. If a table satisfies uniform local residual bounds

$$\underline{V}_t \geq \mathcal{T}_t \underline{V}_{t+1} - \tau_t^-, \quad \bar{V}_t \leq \mathcal{T}_t \bar{V}_{t+1} + \tau_t^+,$$

then its width is at most $\tau_t^- + \tau_t^+ + \beta \max_j e_{t+1}(j)$. Repeating this gives the discounted sum of local residuals. This is a conditional residual theorem, not a claim that a chosen mesh automatically has small residuals. Our actual numerical error certificate instead uses main equation (39), which is directly checked for each stored triangle.

There is a useful distinction between the two vertex calculations. If L_{C_0} is the affine interpolant on a triangle, then for every plane H_k ,

$$\sup_{C_0} (\bar{V} - \underline{V}) \leq \sup_{C_0} (H_k - L_{C_0}) = \max_{i \in \text{vert}(C_0)} (H_k(q_i, b_i) - l_i).$$

Minimizing the right side gives a valid bound. In contrast, interchanging the minimum over planes and the maximum over vertices can underestimate an interior gap where different planes meet. The verifier recomputes the correct min–max order with exact fractions.

If an old and a new collection have been validated, their union remains valid. Keeping all old anchors can only raise the lower envelope; keeping all old planes can only lower the upper envelope. This gives monotone *certified* refinement even when a fresh numerical solve or a separately generated mesh is not itself monotone. The rectangular triangle bound is conservative and need not decrease under arbitrary changes of triangulation; the actual envelope width does decrease under the stated union. We make no problem-independent grid-size or bit-complexity claim.

EC.11.3 Proof of the omitted-coordinate examples

In the promise example, write terminal actions u, v and date-1 actions l, h . The root promise and caps imply

$$2 = x_0 + \frac{1}{2}(l + u + h + v) \leq 1 + \frac{1}{2}(3/4 + 5/4) = 2.$$

Consequently $x_0 = 1$, $l + u = 3/4$, and $h + v = 5/4$. The full optimum equates the two marginal rewards in each branch: $1/4 - l = -u$ and $-1/4 - h = -v$. These give $l = h = 1/2$, $u = 1/4$, $v = 3/4$, with value $-25/32$. If a current-regime table requires $u = v = c$, then $l = 3/4 - c$, $h = 5/4 - c$, and the box

requires $c \in [1/4, 3/4]$. Its value is $-35/32 + c - c^2$, uniquely maximized at $c = 1/2$, giving $-27/32$. The root probability is 1 and every nonroot probability is $1/2$. Weighted projection of the full terminal pair $(1/4, 3/4)$ onto their common-action line has squared Q -distance $1/16$, hence the lower bound $1/32$. This is an all-public, Markov-cap example. Under the unique full optimizer both terminal histories have inherited tier $1/2$ and the same regime; their different exact promises determine different terminal actions. A state omitting the promise cannot implement that optimum.

In the inherited-tier example, the payment equation fixes the second action at $1 - x$. Conditional cost is

$$c_q(x) = \frac{1}{2}[x^2 + (1 - x)^2] + \frac{3}{2}|x - q| + \frac{1}{2}|1 - 2x|.$$

For $q = 0$ and $x \leq 1/2$, it is $x^2 - x/2 + 1$, uniquely minimized at $1/4$. On $[1/2, 1]$ its derivative is $2x + 3/2 > 0$. Symmetry gives $3/4$ for $q = 1$ and cost $15/16$ in both cases. Under a common action, the average incoming switching cost is $3/4$ and the remaining cost is minimized at $x = 1/2$, giving total 1. Here the inherited tiers are public initial conditions of two otherwise identical continuation problems with the same remaining promise. This example isolates the incoming-tier restriction; it is not relabeled as the five-node promise example.

For the inexact version of main Proposition C.3, the scalar expression is evaluated outward: use a lower bound on the projected distance and an upper bound on $\sqrt{2\epsilon}$. A nonpositive result does not establish zero information value. Positive within-cell variation alone is insufficient when the implemented full-class policy has a large certified optimization gap. The argument provides a sufficient lower bound, not an exact decomposition of every comparator hierarchy.

EC.11.4 Continuous-state experiment and independent replay

The new experiment has two regimes $z = 0, 1$, transition rows $(3/4, 1/4)$ and $(1/4, 3/4)$, discount $\beta = 3/4$, and the following rational primitives:

$$a_t(z) = d_t(z) = 1, \quad r_t(z) = (2 + 2z)/3 + (t \bmod 3)/10, \quad \ell_t(z) = (1 + z)/8, \quad \bar{b}_t(z) = \frac{3 + z}{5} \sum_{k=0}^{T-t-1} \beta^k.$$

We use horizons and mesh subdivisions $(T, m) = (3, 2), (4, 4), (4, 8), (4, 16), (8, 4), (16, 4)$. The anchors are the complete grid $q = i/m, b = jU_t(z)/m, 0 \leq i, j \leq m$. A parent lower proposal is a linear program over convex weights on stage-action knots and child anchors, together with the exact promise and inherited-tier equations. Linear interpolation of the concave stage quadratic supplies a lower stage value during proposal generation. The stored witness uses the actual quadratic at its resulting deterministic action, which is at least that chord value. An exact rational solve on the proposed basic support repairs equality rounding before the witness is accepted.

An upper proposal uses tangents to the stage quadratic and the already validated child planes in a linear program. Its numerical dual suggests s, η , and child-plane weights; it does not certify them. Switching prices are clipped to their valid interval, child weights are rounded nonnegative and normalized exactly, and main equation (37) is recomputed rationally. Anchor values are rounded downward and plane constants upward to

a multiple of 10^{-12} . These one-sided changes preserve induction and prevent denominator growth in stored continuation values. All numerical coefficients, mixtures, planes, and interval endpoints are stored.

The separate verifier imports only the Python standard library. It checks every probability row, exact domain, promise equation, inherited-tier equation, mixture, lower reward, switching-price cap, quadratic/interval support constant, cross-envelope anchor inequality, and triangle certificate. It also recomputes the exact information examples. Eleven negative controls alter discounts, caps, mixtures, promises, lower rewards, dual capacities, upper constants, terminal domains, or triangle bounds, or attempt deterministic interpolation of indivisible actions. All are rejected. The six models contain 4,414 certified lower anchors and 1,123 recursive upper planes. These are finite rational *witnesses for the stated continuum inequalities*; they are not exact numerical optimizer labels or a proof that an untested stochastic model has the same approximation error.

No wall-clock speed comparison with a history-tree optimizer is made. The sixteen-period history-tree count in the table is the hypothetical complete binary-tree size; the algorithm does not materialize it. The older finite-action enumeration, hierarchy optimizations, robust diagnostics, and matched learned timings remain unchanged and separately labeled. In particular, the new continuous-state test does not turn the earlier negative learning comparison into a positive result.

EC.12 Correlated Comparator Prices, Reuse Loss, and Uniform Certificates

EC.12.1 Derivation and scope of the reusable certificate

The notation in this section is that of main equations (50)–(52). The diagonal-plus-resource decomposition is $Q = H + R^\top R$, not a removal of the coupled quadratic cost. The equality matrix F_θ includes the actual payment commitments and the chosen information restriction. A changed equality must be evaluated with its actual coefficients and right-hand side.

For any x , the resource inequality

$$-\frac{1}{2} \|Rx\|^2 \leq -\alpha^\top Rx + \frac{1}{2} \|\alpha\|^2$$

has gap $\|\alpha - Rx\|^2/2$. The switching inequality $-\lambda_\theta \kappa_j |u| \leq -\lambda_\theta \tau_j u$ has gap $\lambda_\theta (\kappa_j |u| - \tau_j u)$, nonnegative for $|\tau_j| \leq \kappa_j$. For feasible x , adding $\mu^\top (b_\theta - A_\theta x)$ and $\nu^\top (f_\theta - F_\theta x)$ gives an upper bound because the first term is nonnegative and the second is zero. The remaining coordinate maximizations have the clipped maximizers stated in the main text. This proves the bound and the exact four-term decomposition without a constraint qualification. It also shows why the offset c_θ and the term $\lambda_\theta \tau^\top h$ must not be omitted.

At an anchor with $\lambda_i > 0$, an audited tension s_i is stored as $\tau_i = s_i/\lambda_i$. At a zero-friction anchor the audited tension must be zero, and storing $\tau_i = 0$ is globally safe. Keeping the old unscaled s_i when friction decreases is not safe in general: its magnitude may exceed the new switching capacity. Neither normalizing the tension nor weak duality assumes that the numerical anchor is an exact optimizer. The local loss statement separately requires an exact primal–dual anchor and a retained priced face.

The certificate also remains valid after changing $A_\theta, F_\theta, b_\theta, f_\theta, p_\theta, \lambda_\theta, c_\theta$ simultaneously, provided

the dimensions agree and the displayed common box and cost matrices remain fixed. No common outer polyhedron replaces the model-specific continuation rows. This preserves correlations in the evaluated coefficients, not an inferred statistical correlation in field data. An empty restricted class is excluded by the model, not interpreted as an infinite economic gain.

If these coefficients depend affinely on a d -vector, precompute the constant and d coefficient vectors of $d_\theta(a)$ and of the scalar affine terms for each anchor. The query then requires order $n(d + 1)$ operations and order $n(d + 1)$ stored coefficients per anchor, followed by n scalar clips. Without this preprocessing, matrix–vector products are charged at their sparse sizes. This complexity counts arithmetic, not rational bit operations, cache construction, implemented-policy feasibility, or possible refresh solves. The deployed function receives only query coefficients and cached prices; it does not receive a query comparator optimizer or its value.

EC.12.2 Explicit local constants and the active-face boundary

For the exact-anchor part of Theorem D.3, define $\delta = \|\theta - \theta_0\|$. Suppose throughout a specified neighborhood that

$$\|z_\theta - z_0\| \leq L_z \delta, \quad \|p_\theta - p_0\| \leq L_p \delta, \quad \|A_\theta - A_0\| \leq L_A \delta, \quad \|F_\theta - F_0\| \leq L_F \delta, \quad |\lambda_\theta - \lambda_0| \leq L_\lambda \delta.$$

Here matrix norms are Euclidean operator norms. Because $d_0 - H z_0 - v_0 = 0$,

$$e_\theta = (p_\theta - p_0) - (A_\theta - A_0)^\top \mu - (F_\theta - F_0)^\top v - (\lambda_\theta - \lambda_0) D^\top \tau - H(z_\theta - z_0).$$

Consequently, for $h_{\min} = \min_i H_{ii} > 0$ and $C = L_p + L_A \|\mu\| + L_F \|v\| + L_\lambda \|D^\top \tau\|$,

$$\mathcal{U}_\theta(a) - K_\theta \leq \frac{1}{2} \left\{ \|R\|^2 L_z^2 + \frac{(C + \|H\| L_z)^2}{h_{\min}} \right\} \delta^2. \quad (4)$$

The constants refer to a verified neighborhood, not to an unspecified asymptotic limit. Smooth coefficients, an independent retained active constraint system, strict complementary inequalities and box faces, and interior tensions on retained zero-switch edges give one sufficient regularity route: the nonsingular active KKT system yields a locally differentiable optimizer, and strict inequalities preserve the face. This is a sufficient route, not an assertion that the numerical study certifies such a neighborhood for every instance. The main inequality itself only uses its explicit priced-face conditions.

There is no contradiction between quadratic reuse loss here and the linear local bound for a common outer comparator. The two bounds discard different information. The outer construction relaxes participation rows; the normalized cache retains the actual rows. Nor does the quadratic bound assert that every fixed anchor is better than the outer comparator. Its exact-anchor error is zero, whereas an approximate anchor can have a nonzero residual at its center. Formula (53) remains the appropriate identity in that case.

An exact moving-geometry example has one tier $x \in [0, 1]$, reward $p_t = 2 + 2t$, friction $\lambda_t = 1/2 + t/4$,

cost $x^2/2 + \lambda_t|x|$, and participation row

$$(1+t)x \leq \frac{1}{2} + \frac{1}{8}t, \quad -\frac{1}{4} \leq t \leq \frac{1}{4}.$$

The optimizer is $z_t = (1/2 + t/8)/(1+t)$. Its participation multiplier is positive. At zero, $z_0 = 1/2$ and the exact normalized price is $\mu = \tau = 1$. Its query residual is $d_t = 1/2 + 3t/4$, which remains interior to the tier box. Therefore

$$\mathcal{U}_t - K_t = \frac{1}{2} \left(\frac{t(9/8 + 3t/4)}{1+t} \right)^2.$$

All three changes—reward, participation coefficients and budget, and switching friction—are retained in this identity. The verifier checks it at 17 rational parameters in addition to the algebraic proof. The degenerate two-cap example in the main text is checked at 17 parameters as well; it is a deliberate failure of an unconditional quadratic claim, not an exception omitted from the evidence.

EC.12.3 Uniform uncertainty: the order of the extrema matters

For an affine coefficient family on a polytope, fixed-price upper functions are convex even when their active scalar clips change. Thus their maxima, and the minima of fixed-policy gain minus those functions, are bounded by vertices. This observation proves Corollary D.4 without differentiability, an unchanged active set, or an interior promise.

To verify subdivision, fix an old cell and its selected anchor. Every new vertex in that cell is a convex combination of old vertices, so the concave gain lower function at that vertex is at least the old minimum. Every subcell can retain that anchor or choose a better one. Taking the minimum over subcells cannot worsen the old lower certificate. Adding prices is similarly monotone because the previous maximizing choice remains available. These are finite certificates; increasing dimension or the number of cells can still make vertex enumeration expensive.

The order of extrema cannot be reversed. Consider $0 \leq x \leq 1$, payoff $x - x^2/2$, and rows $x \leq 1+t$, $x \leq 1-t$ on $t \in [-1, 1]$. The two dual prices $(1, 0)$ and $(0, 1)$ give valid affine upper bounds $1+t$ and $1-t$. Their pointwise minimum is zero at both endpoints, while the true value at $t = 0$ is $1/2$. Thus the maximum over vertices of the minimum over anchors is not a uniform upper certificate. For a gain certificate the analogous mistake is selecting an anchor separately at each vertex before taking the vertex minimum. The verifier deliberately attempts the invalid uniform bound and rejects it.

For a positive uniform economic certificate, consider the same scalar reward and friction as in the preceding moving-geometry example, but use the participation row $(1+t)x \leq 3/4 + t/8$ and restrict the comparator to $x = 1/2$. The implemented tier $\hat{x} = 3/5$ and the comparator are both feasible for every $t \in [-1/4, 1/4]$. The restricted optimum is exactly that singleton. The implemented gain is

$$W_t(3/5) - K_t = \frac{19}{200} + \frac{7}{40}t, \quad \inf_t \{W_t(3/5) - K_t\} = \frac{41}{800}.$$

At anchors $a = -1/4, 0, 1/4$, use $\mu = 0, \tau = 1$, and equality price $v_a = 1 + 7a/4$. The cached upper bound is

$\mathcal{U}_t(a) = v_a/2 + \psi(1/2 + 7(t - a)/4)$, with the box clip retained. The exact cell certificates are

equal cells	1	2	4	8	16
uniform gain lower bound	$-569/12800$	$-9/12800$	$2519/51200$	$2519/51200$	$2519/51200$

The four-cell lower endpoint is 0.04919921875, below the true worst gain 0.05125. Further subdivision does not remove the approximation loss of a fixed finite cache. The example establishes a continuous, positive gain certificate, not a rate of convergence for arbitrary cache or cell refinements.

EC.12.4 New and inherited evidence: exact specification

The joint coefficient family is rebuilt from the R25 rational primitives: 63 history nodes, two service coordinates, six coupled resource factors, a positive diagonal curvature, 124 nonroot service-continuation rows, four shared capacity rows, and two root equalities. The time-only comparator additionally pools each service’s amendment at histories with equal date. All numerical comparisons below use this optimized restricted class; a full-class candidate is separately checked against the root equalities and all accepted constraints.

For context c and uncertainty coordinates $\vartheta \in (-1, 1)^3$, the radius- ρ family has

$$p_i = (1 + \rho\vartheta_1)p_i^0 + \rho\vartheta_3(-1)^i/8, \quad \lambda = (1 + \rho\vartheta_2)\lambda_0, \quad A_{ji} = A_{ji}^0(1 + \rho\vartheta_3\sigma_i),$$

where σ_i alternates by history node, shared capacities scale by $1 + \rho\vartheta_2$, and the root and time-only equalities are unchanged. The historical outer model uses nominal continuation coefficients, the rowwise box support expansion, and capacity scaling $1 + \rho$. The reward and friction coefficients are identical between a true model and its outer comparison. The known gain-relative constant is $\lambda \sum_j \kappa_j |h_j|$; it is included in every certificate.

We fix the first eight R25 contexts and retain three independently audited historical comparator prices in each: radii 0, 1/16, and 1/4, along that context’s recorded coefficient direction. Their tension is normalized before storage. The other seven recorded radii per context give 56 holdouts whose optimizer information is not supplied to the deployed cache. These are new evaluations of inherited optimization records, not 56 new solves or a prospectively registered validation sample.

A second study generates 24 new off-ray queries: for each context $i = 0, \dots, 7$ and $k = 0, 1, 2$, set $\rho = 1/64, 1/16, 1/8$, respectively, and take the three integer numerators

$$((-5 + 3i + 2k) \bmod 13) - 6, \quad ((4 + 2i + 5k) \bmod 13) - 6, \quad ((7 + 4i + 3k) \bmod 13) - 6,$$

divided by eight. If the resulting direction is collinear with the inherited one, increment the first coordinate that breaks collinearity, cyclically in $\{-6, \dots, 6\}$. This exclusion depends only on coefficients, not on a measured value or algorithm outcome. The complete realized design is stored before its proposal solves. It is a structured sensitivity study, not a random field sample. Its prices remain the same 24 historical anchors; no new-query optimizer is added to the cache.

For each new query, the proposer solves the true restricted problem, its historical outer relaxation, and

the full accepted problem, for 72 new proposals. It uses the inherited OSQP adapter and rational primal repair. The independent R27 verifier imports none of those routines. It reconstructs every model coefficient, every root and pooling equality, the tier box, and all participation rows. It independently recomputes the scalar conjugates, normalized tension capacities, and each primal–dual bracket. It checks 160 inherited and 72 new optimization records, and 240 cached-price evaluations on the 80 non-anchor queries. Every cache gap at the offline restricted witness is decomposed exactly into its four nonnegative terms. The new full-class policies are feasible proposals, not exact optimizer labels.

The table entries are means of $U - L_K$, where L_K is the independently checked restricted feasible value; the true cache loss is enclosed by $[\max\{0, U - U_K\}, U - L_K]$. Every displayed upper decimal is rounded upward. The largest optimization bracket across the audited records is below 6×10^{-8} . For the 24 new queries, all cache economic-gain certificates are positive and all are tighter than the matched outer certificates. This finite-design statement does not imply that a finite cache dominates every outer construction or is faster in complete deployment.

All initial optimizer work is recorded, and the cache’s query routine performs no restricted solve. The new 72 proposal solves are offline quality evaluation, not hidden online work. Historical solver seconds, new solver seconds, and rational verification are separated in the execution record; no unmatched timing ratio is advertised. The 14 rejection tests cover corrupted coefficients, promises, multiplier signs and dimensions, unscaled tensions, objective endpoints, a false unconditional quadratic rate, and the invalid vertex/anchor interchange. Their role is to test the implementation of the stated mathematics, not to measure scientific novelty.

EC.13 The Shared-Table Bridge: Complete Proof and Representation Costs

EC.13.1 Institution and quantifier order

Let the finite-horizon exogenous Markov process have positive transition probabilities on its retained edges. Its finite public memory updates deterministically. A history h at date t has cell $c(h) = (t, m(h))$ and discounted reach probability $w_h > 0$. Tier and table coordinates lie in $[0, 1]$. A global table $u \in [0, 1]^K$ is chosen at the root; every occurrence of a cell uses the same coordinate. Continuation caps and coefficients have the Markov description stated in the main text. Fix an exact feasible root promise. At a state reached under a fixed table the inherited tier and the conditional remaining payment are (q, b) .

The order $\max_u \mathcal{V}_0(\cdot; u, \delta)$ is essential. In particular, $\max_u \sum_j p_j \mathcal{V}_j(\cdot; u, \delta)$ cannot in general be replaced by $\sum_j p_j \max_{u^j} \mathcal{V}_j(\cdot; u^j, \delta)$. Doing so changes a common contractual table into separate contingent tables. It removes the very restriction being valued.

For fixed u, δ , the admissible interval at cell c is

$$l_c = \max(0, u_c - \delta \omega_c), \quad u_c^{\text{act}} = \min(1, u_c + \delta \omega_c).$$

With positive scalar payment coefficients, the exact feasible promise intervals are obtained backward by

$$L_t = a_t l_c + \beta \sum_j p_j L_{t+1,j}, \quad U_t = \min\{\bar{b}_t, a_t u_c^{\text{act}} + \beta \sum_j p_j U_{t+1,j}\}.$$

Terminal intervals are $\{0\}$. An empty child interval or $U_t < L_t$ gives an empty parent domain. All intermediate payments are attainable: sums of intervals are intervals, and actions and child promises vary continuously. Feasibility does not depend on the inherited tier in this institution. These intervals can be used for fixed-table lower witnesses. The affine upper-plane recursion instead uses the *unrestricted* intervals, obtained with action interval $[0, 1]$, because its coefficients must remain valid as u varies.

Given an accepted history policy, assign at each child its actual conditional discounted remaining payment. The tower property gives the parent promise equality. Conditional optimization at fixed u can only improve this policy. Conversely, an attained recursion choice and attained successor policies concatenate into a history policy with all required payments and caps. Induction proves equality for each fixed table. Compactness of the original finite-dimensional feasible graph gives an attained optimum over u whenever the root problem is feasible. Maximizing only once proves the first assertion of Theorem C.4.

For a scalar cell, the intervals $[x_h - \delta\omega_c, x_h + \delta\omega_c]$ have a common intersection if and only if $\max_h x_h - \min_h x_h \leq 2\delta\omega_c$. Their midpoint lies in $[0, 1]$ because all tiers do. Thus eliminating u_c gives all pairwise inequalities in the main text, with no extra restriction from the table box. The reduced tier problem has a positive-definite diagonal operating quadratic, a convex absolute-value switching penalty, and a fixed polyhedron with an affine right-side release parameter. The finite-pattern argument for Theorem A.1 applies. Concavity follows also by mixing both the tiers and their tables. No unique table is asserted: the optimized tier vector is unique, but several tables can implement it.

EC.13.2 Validity of recursive prices

Suppose the successor plane inequalities hold throughout their feasible domains. Substitute $q_j = x$ and the common table in these inequalities. The payment equality permits adding $\eta(b - ax - \beta \sum_j p_j b_j)$. Add the nonnegative quantities

$$\chi(\bar{b} - b), \quad \rho^+(u_c + \delta\omega_c - x), \quad \rho^-(x - u_c + \delta\omega_c),$$

and use $-\ell|x - q| \leq -s(x - q)$ for $|s| \leq \ell$. The coefficient of x is d in (46); the coefficient of each child promise is $\beta p_j(B_j - \eta)$. Maximizing the scalar quadratic over the tier box and each child promise over its unrestricted interval gives exactly that equation. These maximizations can enlarge the feasible set, which is permitted for an upper bound. Induction starts with the zero terminal plane. Convex combinations of valid child planes are valid, so they are also allowed by the construction.

For any fixed table, the value is below every valid root plane and hence their minimum. Maximizing the minimum over the same global table gives the epigraph formulation (44), which is a bounded linear program when at least one plane is present. Every accepted state policy implements a lower witness only if its table is shared across all successors; its conditional payments must be actual promised payments, not

interpolated forecasts. These observations prove both sides of (45). Neither weak duality nor a numerical root-LP success code certifies the lower witness by itself.

EC.13.3 Dual completeness uses nested continuation constraints

We give the normalization explicitly. In the finite tree problem, let ν be the multiplier of the exact root payment, and let $\lambda_n \geq 0$ price the *conditional* cap at nonroot node n . Set

$$\chi_n = \lambda_n / w_n, \quad \eta_n = \nu + \sum_{v \leq n, v \neq 0} \chi_v, \quad \chi_0 = 0, \quad \eta_0 = \nu.$$

The multiplier of a conditional cap contributes $w_i a_i \chi_n$ to the payment coefficient of each descendant tier i . This follows from $w_i / w_n = \beta^{t(i)-t(n)} \Pr(i \mid n)$. Therefore the local payment coefficient after normalization is $a_i \eta_i$, and on every edge

$$\eta_j - \chi_j = \eta_n. \quad (5)$$

This identity would not follow from an arbitrary collection of polyhedral rows.

Take an optimum of the joint (x, u) problem. The negative objective is a finite continuous convex function on a polyhedron. Polyhedral normal cones and the subdifferential sum rule supply multipliers, including switching subgradients, without requiring a strict interior feasible point. Divide each node's corridor multipliers by w_n to obtain $\rho_n^\pm$, and normalize its switching tension to $s_n \in [-\ell_n, \ell_n]$. The stage stationarity condition, including incoming switching tension and outgoing child tensions, says exactly that x_n maximizes the local box quadratic with coefficient d_n in (46).

Construct a child plane for every node backward from these multipliers. Its payment coefficient is $B_j = \eta_j - \chi_j = \eta_n$. Thus the child-promise support term is zero and is tight at the actual child promise. Complementarity makes all priced continuation and corridor slacks zero. Switching tensions are exact subgradients, and local box maximization is attained. Induction shows that the resulting root plane equals the optimized policy value at (q_0, b_0, u^*, δ) . Its table coefficient is

$$\gamma_c = \sum_{n:c(n)=c} w_n (\rho_n^+ - \rho_n^-).$$

Stationarity for the *one global* table gives $\gamma \in N_{[0,1]^\kappa}(u^*)$, so u^* maximizes $\gamma^\top u$ over the table box. Consequently the maximum of this single root plane over all tables is exactly $V_M(\delta)$. The entire recursive plane class is therefore dual-complete for each finite problem.

The construction proves existence using tree multipliers; it does *not* propose enumerating that tree as the query algorithm. One may store multiple planes at the same (t, z, m) , reusing identical successor planes as a directed acyclic graph. The theorem does not guarantee that a single price function of the primal state suffices at degeneracy, or that the number of distinct planes in an exact certificate is polynomial in horizon. Finite stored families give valid bounds without either assertion. A master LP certificate itself may be checked through a primal–dual pair; merely maximizing each plane separately is a valid but potentially weaker upper bound.

Unrolling the coefficient recursion yields

$$k_0 = \sum_n w_n \omega_{c(n)} (\rho_n^+ + \rho_n^-).$$

An exact root plane maximized at u^* supports the concave optimized value as the release parameter changes, because its coefficient data remain fixed and its table maximum is an affine function of that parameter. Hence $V_M(\delta') \leq V_M(\delta) + k_0(\delta' - \delta)$ for every feasible release. At a differentiability point this gives the derivative. At a kink, a supporting price is not automatically the selected one-sided derivative: the directional price is the extremum over the optimal dual set specified in Section B. Our exact example reports left and right slopes separately at events and never interprets an arbitrary solver multiplier as a unique directional rent.

EC.13.4 What is compressed, and what accuracy costs

State sufficiency, existence of certified approximations, and complexity are distinct statements. The exact state has finite public coordinates and continuous (q, b) ; additional additive commitments add continuous coordinates. The shared table contributes K ex ante design variables. The state representation eliminates redundant histories, not continuous approximation or design optimization.

For a stored family with at most P_0 planes at each of TS_0M_0 public date–regime–memory states, dense coefficient storage for the table-augmented planes is $O(TS_0M_0P_0(K + 1))$, plus successor references. Verifying each plane from at most B successors costs $O(B(K + 1))$ arithmetic operations for scalar tiers. The root master has $K + 1$ variables and P_0 inequalities besides table bounds. A policy’s lower-envelope storage and child-allocation work are additional; the latter can grow exponentially in branching when combinations are enumerated. Stored witnesses must cover the promised domain on which a uniform claim is made. Evaluating only one root state is not a uniform-domain test. Rational bit complexity is separate from arithmetic counts.

Proposition EC.13.1 (Conditional accuracy rates, not dimension-free rates). *Let a compact polyhedral feasible domain of dimension d_s admit a simplicial mesh of maximum diameter h . Suppose the joint value function is concave and L -Lipschitz there; at every mesh vertex an accepted policy gives value within ε_L of the optimum, and a valid supporting affine upper plane differs from the value at its anchor by at most ε_U and has slope norm at most L . Then deterministic barycentric policy mixtures and the minimum of these planes have width at most*

$$2Lh + \varepsilon_L + \varepsilon_U.$$

If the value has M -Lipschitz gradient on the domain and exact-gradient supporting planes are available with the stated intercept error, the width is at most $Mh^2 + \varepsilon_L + \varepsilon_U$.

Proof. At any point of a simplex, each vertex value differs from the point value by at most Lh . Concavity and feasibility of deterministic mixing give a lower value within $Lh + \varepsilon_L$. A plane from any simplex vertex is at most $2Lh + \varepsilon_U$ above the point value if used alone; a sharper joint estimate uses the barycentric average of the vertex planes. The minimum of the planes is no larger than that average. Subtracting the mixed lower value cancels the vertex values and leaves $\sum_i \lambda_i g_i^\top (y - y_i) + \varepsilon_L + \varepsilon_U \leq Lh + \varepsilon_L + \varepsilon_U$, which in particular

implies the displayed conservative $2Lh$ bound. For the smooth case, the lower interpolation error is at most $Mh^2/2 + \varepsilon_L$ by the descent inequality and barycentric cancellation at the point. The averaged upper error is at most $Mh^2/2 + \varepsilon_U$ by the same inequality at the vertices. Their sum proves the assertion. All mixtures use the joint parameter/state coordinates and their compatible shared tables. $\square$

For a fixed regular domain, covering by $O(h^{-d_s})$ simplices gives conditional counts $O((L/\varepsilon)^{d_s})$, or $O((M/\varepsilon)^{d_s/2})$ in the smooth case, apart from anchor accuracy and geometric constants. A mesh joint in (q, b, u, δ) includes the table dimension; treating the table by the master problem does not require such a mesh but does require certified planes. Nonsmooth active-set transitions generally invalidate the smooth rate. Boundary promises require the domain-relative Lipschitz and support assumptions just stated; they do not follow from compactness alone. Without these assumptions the earlier a posteriori gap remains valid, but no rate or finite mesh size for a requested tolerance is claimed.

EC.14 Exact Bridge Example and an Endogenous Release Decision

The two-period instance in Section C.3 has objective

$$W(x) = x_0 - x_0^2/2 + x_H - (x_L^2 + x_H^2)/4, \quad x_0 + (x_L + x_H)/2 = 1.$$

The low and high child caps are $2/5$ and $4/5$. Eliminating the shared child table gives $|x_H - x_L| \leq 2\delta$. The complete optimized policy is

$$(x_0, x_L, x_H) = \begin{cases} (3/5 - \delta, 2/5, 2/5 + 2\delta), & 0 \leq \delta \leq 1/10, \\ (1/2, 1/2 - \delta, 1/2 + \delta), & 1/10 \leq \delta \leq 3/10, \\ (1/5 + \delta, 4/5 - 2\delta, 4/5), & 3/10 \leq \delta \leq 2/5, \\ (3/5, 0, 4/5), & \delta \geq 2/5. \end{cases}$$

Substitution gives

$$V_M(\delta) = \begin{cases} 37/50 + 6\delta/5 - 3\delta^2/2, & 0 \leq \delta \leq 1/10, \\ 3/4 + \delta - \delta^2/2, & 1/10 \leq \delta \leq 3/10, \\ 33/50 + 8\delta/5 - 3\delta^2/2, & 3/10 \leq \delta \leq 2/5, \\ 53/50, & \delta \geq 2/5. \end{cases}$$

All segments agree at their endpoints. Let t price $x_H - x_L \leq 2\delta$, and let μ_L, μ_H price the two conditional child caps in the global objective normalization. Exact witnesses on the three nonconstant open segments are respectively

$$(\eta, t, \mu_L, \mu_H) = \begin{cases} (2/5 + \delta, 3/5 - 3\delta/2, 1/5 - 2\delta, 0), \\ (1/2, 1/2 - \delta/2, 0, 0), \\ (4/5 - \delta, 4/5 - 3\delta/2, 0, 2\delta - 3/5). \end{cases}$$

All are nonnegative where required. The normalized child participation prices are $\chi_j = 2\mu_j$. Use corridor prices $\rho_H^+ = 2t$, $\rho_L^- = 2t$, and all other corridor prices zero; the common child table is $(x_L + x_H)/2$. The child payment prices are $\eta_j = \eta + 2\mu_j$. The two child table coefficients cancel in the root plane, while its release coefficient is $2t$. Local box conjugates are attained at the displayed policy. Hence the recursively constructed plane is an exact upper bound on the optimized table, not merely on a fixed witness. On the constant final segment use $t = 0$, $\mu_H = 2/5$, $\mu_L = 0$, and $\eta = 2/5$; the low child is at its tier lower bound.

At $\delta = 2/5$ the left derivative is $2/5$ and the right derivative is zero. Both are supported by optimal dual witnesses; a selected price between them is not a unique directional derivative. At zero release, the forward restriction price above realizes the right derivative. The replay checks stationarity, complementarity, local scalar maximization, recursive price equality, and all-policy support at every declared rational test release.

For the design cost $C(\delta) = \delta$, the first-segment derivative equation $6/5 - 3\delta = 1$ yields $\delta = 1/15$. Concavity proves global net optimality. Its gross value is $61/75$, net value is $56/75$, and net improvement over no release is $1/150$. Full release has higher gross value but lower net value under this cost. These are stylized decision inputs, not calibrated installation costs.

EC.15 Computable Cache Governance and Complete Query Accounting

Let z^w be a feasible restricted witness, not necessarily optimal. Weak duality gives $W_\theta(z^w) \leq K_\theta \leq U$. Subtraction proves Proposition D.5. Because every term in the four-term decomposition of Theorem D.3 is nonnegative at a feasible witness, the width also identifies how much is charged to resource-price mismatch, coordinate response, participation slack, and switching mismatch. It is an upper bound on cache error, not an equality with that error. A poor witness can cause an unnecessary refresh; it cannot create a false quality certificate.

An implementation must first audit the witness against the *actual* query coefficients. If the witness is infeasible, the width is not certified and the controller must repair, refresh, or abstain. With a feasible witness, return the cache only if $U - W_\theta(z^w) \leq \varepsilon$. Otherwise obtain a new independently checked primal–dual pair, or return “tolerance uncertified.” The test never converts a failure into a proof of infeasibility or poor optimal value. A complete economic decision separately checks the implemented full-class policy, evaluates its actual objective, subtracts U , and subtracts any applicable implementation cost. Thus a good comparator cache alone does not authorize deployment.

EC.15.1 An optional sufficient face-retention test

Select a linearly independent equality basis $B_\theta z = c_\theta$ consisting of the mandatory equalities, proposed active inequality or box rows, and proposed zero-switching rows. Specify the signs σ_j of all remaining switching rows, collected in D_N . Solve

$$\begin{bmatrix} Q & B_\theta^\top \\ B_\theta & 0 \end{bmatrix} \begin{bmatrix} z \\ \zeta \end{bmatrix} = \begin{bmatrix} p_\theta - \lambda_\theta D_N^\top (\kappa_N \sigma_N) \\ c_\theta \end{bmatrix}. \quad (6)$$

A full-row-rank B_θ and $Q \succ 0$ make this matrix nonsingular. Set the prices of unselected inequalities to zero. Check all original inequalities and equalities, every box constraint, the proposed nonzero switching signs, nonnegativity of selected inequality prices in their original orientations, and $|\zeta_j| \leq \lambda_\theta \kappa_j$ for the zero-switching tensions. Check complementarity and stationarity in the original coordinates. If the representation is valid, Proposition 3.1 certifies that z is the unique query restricted optimum. At a degenerate face the chosen independent basis can fail even when some alternative representation works; the procedure is sufficient, not necessary. A singular or inconsistent selected basis is rejected rather than silently inverted.

For the cached anchor in the local clause of Theorem D.3, additionally check its positive-price complementarity at z , membership of its fixed v_0 in the query box normal, and vanishing of its switching gap. The local upper bound then becomes computable from the verified z . When the coefficient matrix is fixed and right sides/rewards are affine, z is affine on this checked face and all sign and feasibility tests define a polyhedral retained-face region. Its Lipschitz modulus is obtained from the displayed inverse times the parameter loadings. When B_θ itself varies, nonsingularity and a lower singular-value bound on a neighborhood give a local Lipschitz bound, but refactorization and regional verification must be accounted for. No cheap fixed-factorization claim is made for changing matrices.

This test is equality-constrained linear algebra followed by a full certificate audit, not an unqualified replacement for constrained optimization. Face transitions can trigger rejection; the original cached weak-duality upper bound remains valid. In particular the retained R27 two-cap example still has linear cache loss across its degenerate face. The R28 checker includes rejection of an infeasible retained-face proposal and a singular active basis.

EC.15.2 Cost components

For a full-vector model, a complete query budget is

$$C_{\text{coeff}} + C_{\text{candidate}} + C_{\text{audit}} + C_{\text{cache}} + \mathbf{1}_{\text{refresh}}(C_{\text{solve}} + C_{\text{dual audit}} + C_{\text{update}}).$$

Here coefficient construction is distinct from matrix preprocessing. Candidate generation may involve a full-class optimizer and is never called “no query optimization.” Its feasibility audit includes $A_\theta \hat{x}$, $F_\theta \hat{x}$, boxes, switching, and objective evaluation; dense costs can dominate cached arithmetic. With k_a anchors, cached evaluation uses $O(k_a n(d + 1))$ arithmetic after the stated preprocessing, plus storage of all needed coefficients/prices. A witness repair, its audit, and the quality test are additional. A face-KKT test adds a solve with the stored factorization or, when coefficients change, a factorization and its verification. For the state bridge the corresponding charges are policy/promise audits, recursive plane verification, and the shared-table master LP. Neither representation hides these costs in a nominal cache lookup.

EC.16 Uniform Certification of a Parameter-Dependent Policy

On a simplex, write $\theta = \sum_i \lambda_i v_i$ and $\hat{x}(\theta) = \sum_i \lambda_i x_i$, with $\lambda_i \geq 0$ and $\sum_i \lambda_i = 1$. Affine coefficient dependence gives

$$b_\theta - A_\theta \hat{x}(\theta) = \sum_i \lambda_i^2 (b_i - A_i x_i) + 2 \sum_{i < j} \lambda_i \lambda_j \frac{b_i + b_j - A_i x_j - A_j x_i}{2}.$$

The quadratic Bernstein basis weights are nonnegative and sum to one. Nonnegative coefficients therefore certify every inequality throughout the simplex. The same identity proves the accepted equalities $E_\theta \hat{x} = e_\theta$ when all coefficients are zero; comparator-only rows in F_θ are not imposed on the deployed policy. These are sufficient coefficient tests, not a claim that vertex feasibility implies feasibility. The common box is convex, so its vertex test is sufficient. The example $\theta(1 - \theta) \leq 0$ illustrates why the off-diagonal terms cannot be dropped.

After partitioning at the affine switching hyperplanes, each absolute-value term in the deployed objective has one sign, including its zero boundary. Since the friction scale is affine and the quadratic geometry is fixed, $W_\theta(\hat{x}(\theta))$ is a polynomial of degree at most two on each resulting simplex. The R27 upper bound $U_\theta(a)$ is convex in the parameter for a fixed cached price under its fixed-geometry hypotheses. Hence

$$U_\theta(a) \leq \sum_i \lambda_i U_{v_i}(a), \quad W_\theta(\hat{x}(\theta)) - K_\theta \geq f_a(\theta).$$

A quadratic's diagonal Bernstein coefficients are its vertex values. Evaluation at the edge midpoint gives $f_a((v_i + v_j)/2) = (g_{ii,a} + 2g_{ij,a} + g_{jj,a})/4$, proving the coefficient formula. Taking the smallest coefficient is a lower bound on the whole simplex. Any anchor gives a bound throughout that simplex, so the largest of the anchorwise lower bounds remains valid. Taking the smallest bound over cells proves (57). Exchanging these extrema or checking only the best anchor separately at each vertex is not justified.

All restrictions are assumed nonempty on every cell. This can be established by a separate restricted witness map passing the same feasibility test; it is not implied by full-policy feasibility. The deployed map, its partition, and the assignment on cell boundaries are part of the certificate. Query-specific optimizer outputs with no declared interpolation law remain pointwise evidence, including the inherited R27 off-ray study. The R28 example below, unlike that study, certifies an explicitly declared nonconstant map on an entire interval.

EC.16.1 An exact positive adaptive example

Let $\theta \in [1/8, 3/8]$, $x \in [0, 1]^2$, and impose $x_1 + x_2 = 1$ and

$$(1 + \theta)x_1 + (1 - \theta)x_2 \leq 1 + \theta.$$

The restricted comparator additionally requires $x_1 = x_2$. The reward is $p_\theta = (1 + \theta, 1 - \theta)$, $Q = I$, switching operator $D = (1, -1)$, offset zero, weight one, and friction scale $\theta/4$. Thus curvature and switching geometry

stay fixed. The constant restricted witness $(1/2, 1/2)$ proves nonemptiness throughout the interval and has optimized value $K_\theta = 3/4$.

Declare the adaptive policy

$$\hat{x}(\theta) = (1/2 + 3\theta/4, 1/2 - 3\theta/4).$$

It is in the box, satisfies the payment equality, and has cap slack $\theta - 3\theta^2/2 \geq 0$. All coefficients of the corresponding Bernstein slack on the declared interval are nonnegative. Its gain is $W_\theta(\hat{x}(\theta)) - K_\theta = 9\theta^2/16$.

Use the one exact cached anchor at $\theta_0 = 1/8$ with zero cap price, root-payment price $1/2$, pooling price $3\theta_0/4$, and normalized switching tension one. Its actual-query upper bound is

$$U_\theta = 3/4 + 9(\theta - \theta_0)^2/16.$$

For the vertex-chord polynomial, the Bernstein coefficients are $9/1024$, $9/1024$, and $45/1024$. Thus Proposition D.6 certifies the nonconstant policy's gain uniformly by $9/1024 > 0$, attained at the left endpoint. The independent checker reconstructs the prices, objective, feasibility coefficients, chord, and all three gain coefficients with rational arithmetic. It also rejects the endpoint-only feasibility counterexample.

EC.17 Degenerate Frontiers and Closest-Results Comparison

The phrase “complete release frontier” denotes the finite global structure, not a guarantee that successive regular-cell formulas alone constitute a robust continuation algorithm. A complete, generally exponential procedure is available: enumerate every switching sign pattern and every independent active-row basis; solve its equality-constrained strictly concave quadratic; test the original sign constraints and all primal–dual feasibility conditions, allowing zero multipliers; project the resulting parameter polyhedron onto the release line; retain the valid intervals and their common unique tier optimizer. Redundant active rows may be assigned zero prices when a valid independent generating subset exists, or their multiplier feasibility can be retained explicitly as a linear feasibility problem. Simultaneous events and isolated feasible cells must be retained. There are finitely many patterns and subsets, so exhaustive enumeration terminates. Optimality follows from the balance criterion, and completeness from polyhedral normal-cone representation and the finite switching partition. This is a structural finite algorithm, not a polynomial or cheap zero-set method. The friction-path reduction has the same limitation.

The central theorem chain in this revision is shared-table implementation, recursively accumulated continuation/restriction prices, and certified net design value. The following comparison distinguishes inherited engines from the added structural conclusion. The complete closest-result comparison appears in main Table 1; the distinction between an inherited engine and the shared-table conclusion is part of the main argument.

Rockafellar and Wets (1991) already make information bundles and nonanticipativity central to policy aggregation; Benders (1962) already separate shared design variables from subproblems. Pereira and Pinto (1991) already use stochastic dual value approximations. We do not rename those methods as inventions. Their general statements do not by themselves establish that the optimized pooled contract with an exact

continuation obligation is represented by the displayed state recursion, or that its marginal release value and optimized-comparator certificate are the same recursively assembled price object. Theorem C.4 supplies that contract-specific representation and its proof. Its exactness exploits both nested conditional caps and the table's global stationarity. The theorem does not establish priority over all possible combinations of these literatures; its comparison identifies the precise mathematical conclusion proved here.

EC.18 R28 Exact Evidence and Scaling Protocol

EC.18.1 Structural verification

The nine release values are 0, 1/20, 1/15, 1/10, 1/5, 3/10, 7/20, 2/5, 1/2. The generator supplies the displayed policy and normalized prices. The independent verifier does not import the generator. It enumerates independent active bases of the three-dimensional accepted QP, solves each candidate KKT system with fractions, and retains only globally feasible, nonnegative-price solutions. The unique optimal tier vector must match the supplied policy. It then reconstructs the recursive child planes, checks every scalar box conjugate at the policy, verifies the zero shared-table coefficient and exact root upper value, and checks every anchor plane against every declared release. The example's domain-wide certificate is established by the proof, not by these 81 point checks.

A second exact master calculation maximizes the minimum of $u - 1/8$ and $7/8 - u$ over $0 \leq u \leq 1$. These are valid one-period shared-table planes with root promise 1/2, reward and curvature one, and zero release. The maximum 3/8 occurs at $u = 1/2$. This check keeps the table master explicit rather than assuming every plane has zero table coefficient. The adaptive-cell test reconstructs all correlated feasibility and objective coefficients from the declared map in EC Section EC.16. The linear release-cost decision is checked at its stationary release 1/15 and against the exact piecewise-concave frontier.

The fourteen negative controls include: childwise table substitution, a false pooled optimal value, an incorrect release-radius factor, a violated participation witness, incorrect conditional-price normalization, a corrupt recursive release coefficient, endpoint-only adaptive feasibility, a hidden midpoint violation, a stale active-face proposal that violates another cap, a singular active basis, negative cached inequality prices, a broken root promise, an infeasible continuation witness, and an unsupported tolerance declaration. The verifier records each rejection separately. These are finite regression controls, not a formal proof checker for arbitrary mathematical statements.

EC.18.2 Precisely specified stress family

Each independent block has root payment $x_0 + (x_L + x_H)/2 = 1$, tiers in $[0, 1]$, conditional child constraints $a_L x_L \leq b_L$ and $a_H x_H \leq b_H$, and objective

$$p_0 x_0 - x_0^2/2 + (r_L x_L + r_H x_H)/2 - (x_L^2 + x_H^2)/4.$$

The restricted class shares one child tier, $x_L = x_H$. Blocks are independent renewal/service components; a configuration with C cycles and S services has $n = 3CS$ tier variables. Each block has its own exact payment

commitment. This separability makes an exact fresh comparator available and exposes overhead fairly; it is not a dense or long-horizon complexity experiment. There is no new neural model, empirical calibration, or fitted test distribution.

For parameter dimension d and query index $i = 0, \dots, 11$, set

$$\theta_j = \frac{((i+3)(7j+5) + 3j^2) \bmod 19 - 9}{9}, \quad j = 0, \dots, d-1.$$

For block index $b = 0, \dots, CS - 1$, put

$$z = \frac{1}{d} \sum_j \frac{((b+1)(j+2)) \bmod 7 - 3}{3} \theta_j, \quad z' = \frac{1}{d} \sum_j \frac{((b+4)(j+5)) \bmod 9 - 4}{4} \theta_j.$$

Primitives are $p_0 = 1+z/8$, $r_L = 1/5+z'/4$, $r_H = 9/5+z/2$, $a_L = 1+z/8$, and $a_H = 1-z/10$. In the stable-cap design, $b_L = 7/20 + z'/30$ and $b_H = 7/10 - z'/30$. In the unstable-cap design, $b_L = 1/2 + \theta_0/5 + z'/20$ and $b_H = 1/2 - \theta_0/5 - z'/20$. Coefficients and bounds therefore change together, and actual query coefficients enter every certificate. Effective child caps lie strictly between zero and one. The root remains feasible throughout the declared design.

Let $v_f = (1 - p_0 + (r_L + r_H)/2)/2$. The exact restricted tier is

$$v^* = \min\{\max(0, v_f), b_L/a_L, b_H/a_H\}, \quad x^Y = (1 - v^*, v^*, v^*).$$

All measured cases have $0 < v^* < 1$. With $v = p_0 - (1 - v^*)$ and total effective cap price $2(v_f - v^*)$, assign that nonnegative price to a binding smallest cap, dividing by its actual coefficient a_j ; set the other cap price to zero. The pooling price is determined by low-child stationarity. This is an exact primal–dual pair, including at the declared tie convention. It is used as a fresh reference but never interpreted as a unique economic directional derivative at degeneracy. Full policies are generated by exact weighted water filling under the same payment and cap constraints and independently checked through their KKT conditions.

EC.18.3 Governance, timing, and memory

An initial cache of size k_a uses the same parameter formula at indices $100, \dots, 100 + k_a - 1$. The best aggregate upper bound is selected over these global block-price vectors. A restricted witness repairs that anchor’s child tier by clipping it to the actual two child caps and adjusting the root to maintain payment one. This is not reoptimization. If the upper-minus-witness width exceeds $CS/1000$, the controller refreshes with a new audited exact restricted pair and evicts the oldest anchor; otherwise it retains the cache. The cache remains at its declared size. A query’s exact restricted solve is evaluation-only unless the declared trigger refreshes; on a refresh its full measured solve/audit time is charged. The fresh baseline always computes that same exact pair. Both pipelines meet the same requested tolerance, and the exact fresh comparator in fact has zero error.

The independent replay regenerates query primitives, checks all cached dual bounds and their non-negative coordinate/participation decompositions, verifies every implemented policy and restricted witness,

reconstructs each FIFO update, and confirms each trigger and resulting error. It accepts no solver termination labels. It audits 84 aggregate queries and 10,416 query block policies. Performance timing is recorded separately from arithmetic validity.

Each configuration runs in a fresh sequential Python subprocess with standard-library rational arithmetic. Phase times use `perf_counter_ns`; process peak resident memory uses Linux `getrusage`. The complete configuration, Python/platform provenance, raw phase times, prices, policies, and rational bounds are in `results/evidence.json`. Subprocess startup is not charged as a per-query cost. Initial cache construction is reported separately; no amortization advantage is asserted. Serialization bytes describe cache payload size, whereas RSS includes interpreter and working memory. The following tables give phase medians in milliseconds and the observed memory/quality measures; the median of a sum need not equal the sum of the displayed medians.

n	Cache	Coefficients	Policy	Audit	Cache	Witness	Refresh	Fresh solve
12	1	0.14	0.59	0.11	0.21	0.14	0.70	0.50
96	1	1.56	4.66	0.84	1.68	1.07	5.54	3.97
96	4	1.55	4.61	0.85	6.57	1.09	5.41	3.91
96	4	1.69	5.12	0.87	6.52	1.06	5.51	3.95
384	4	8.64	17.83	3.30	25.96	4.11	21.64	15.54
384	8	8.46	17.61	3.24	50.64	4.04	20.99	15.04
1536	8	33.81	70.51	12.82	204.89	15.91	83.22	60.27

n	Cache	Initial build (ms)	Serialized cache (KiB)	Max. stale loss/block	Max. used loss/block
12	1	0.74	0.5	0.08205	0.0000000
96	1	5.34	3.9	0.01404	0.0000000
96	4	20.47	15.5	0.00532	0.0002128
96	4	20.51	15.3	0.02078	0.0000000
384	4	90.50	63.2	0.02586	0.0000000
384	8	179.82	126.9	0.00754	0.0000000
1536	8	710.87	505.6	0.00758	0.0000000

The refresh rule accepts all final comparator certificates at tolerance, but refresh occurs on 83 queries. The negative cost result is not a failure of the upper-bound theorem: it shows that this strict accuracy target and this cache/repair policy frequently require new prices. Neither the global validity theorem nor the local quadratic clause predicts an unconditional speed advantage.

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